

Foundation Programme
Universiti Teknologi Malaysia
Session 2019/2020 Semester 3

Test 1 (15%)

Time: 75 minutes

FSPM 0024 Calculus

Answer all questions.

1. Evaluate the limit for each of the following.

(a) $\lim_{x \rightarrow 0} \frac{\sqrt{x+2} - \sqrt{2}}{x}$. (3 marks)

(b) $\lim_{x \rightarrow \infty} \frac{5x^2 + 2x + 1}{\sqrt{x^4 + 2}}$. (3 marks)

2. Use First Principle Formula to find $f'(x)$ if $f(x) = 2\sqrt{x+10}$. (4 marks)

3. Use appropriate rules to find $\frac{dy}{dx}$ for each of the following.

(a) $y = \operatorname{cosec}(\ln x)$. (3 marks)

(b) $y = \frac{\tan 2x}{e^{x^4}}$. (4 marks)

4. Given $\cos(xy) = 3y^2 - 2e^x$. Using implicit differentiation, find $\frac{dy}{dx}$ at point $(0, 1)$. (4 marks)

5. Given a set of parametric equations

$$x = 2t^2 + 1, \quad \text{and} \quad y = 2\sin(t^2).$$

Find $\frac{dy}{dx}$. Hence, show that $\frac{d^2y}{dx^2} = -\frac{1}{2}\sin(t^2)$. (5 marks)

6. Use differential to find the approximate value of

$$(16.1)^{\frac{1}{4}} + (16.1)^{\frac{1}{2}},$$

accurate to four decimal places.

(4 marks)

$$1) \quad (a) \quad \lim_{x \rightarrow 0} \frac{\sqrt{x+2} - \sqrt{2}}{x} \cdot \frac{\sqrt{x+2} + \sqrt{2}}{\sqrt{x+2} + \sqrt{2}} = \frac{x+2-2}{x(\sqrt{x+2} + \sqrt{2})} = \frac{1}{\sqrt{x+2} + \sqrt{2}} \\ = \frac{1}{\sqrt{2} + \sqrt{2}} = \frac{\sqrt{2}}{4}$$

$$(b) \quad \lim_{x \rightarrow \infty} \frac{5x^2 + 2x + 1}{\sqrt{x^4 + 2}} = \frac{\frac{5x^2}{x^2} + \frac{2x}{x^2} + \frac{1}{x^2}}{\frac{\sqrt{x^4}}{\sqrt{x^4}} + \frac{2}{x^2}} = \frac{5 + \frac{2}{x} + \frac{1}{x^2}}{1 + \frac{2}{x^2}} = 5$$

$$2) \quad f(x) = 2\sqrt{x+10}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \frac{f(x+h) - f(x)}{h} \cdot \frac{2\sqrt{x+h+10} + 2\sqrt{x+10}}{2\sqrt{x+h+10} + 2\sqrt{x+10}} \\ = \frac{(2\sqrt{x+h+10})^2 - (2\sqrt{x+10})^2}{h(2\sqrt{x+h+10} + 2\sqrt{x+10})} \\ = \frac{4x + 4h + 40 - 4x - 40}{h(2\sqrt{x+h+10} + 2\sqrt{x+10})} \\ = \frac{4}{2\sqrt{x+h+10} + 2\sqrt{x+10}} \\ = \frac{4}{4\sqrt{x+10}} \\ = \frac{1}{\sqrt{x+10}}$$

$$3) \quad y = \operatorname{cosec}(\ln x)$$

$$y = \operatorname{cosec} u, \quad u = \ln x, \quad \frac{du}{dx} = \frac{1}{x}$$

$$\frac{dy}{du} = -\operatorname{cosec} u \cot u$$

$$\frac{dy}{dx} = \frac{-\operatorname{cosec}(\ln x) \cot(\ln x)}{x}$$

$$y = \frac{\tan 2x}{e^{x^4}}$$

$$u = \tan 2x$$

$$v = e^{x^4}$$

$$\frac{du}{dx} = 2\sec^2 2x$$

$$\frac{dv}{dx} = e^{x^4} (4x^3)$$

$$\frac{dy}{dx} = \frac{vu' - uv'}{v^2}$$

$$= \frac{e^{x^4} (2\sec^2 2x) - \tan 2x (4x^3)(e^{x^4})}{(e^{x^4})^2}$$

$$= \frac{2\sec^2 2x - 4x^3 \tan 2x}{e^{x^4}}$$

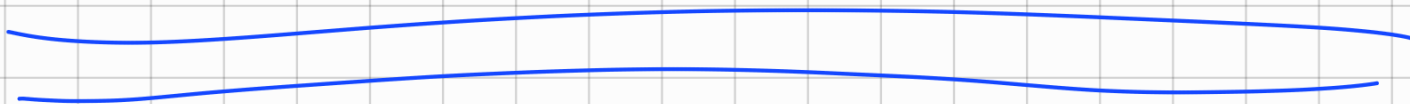
$$= \frac{2 - 2x^3 \sin 4x}{\cos^2 2x (e^{x^4})}$$

$$\begin{aligned}
 4) \quad \cos(xy) &= 3y^2 - 2e^x \\
 \frac{dy}{dx} \cos xy &= \frac{dy}{dx} 3y^2 - \frac{dy}{dx} 2e^x \\
 -\sin xy \left(x \frac{dy}{dx} + y \right) &= 6y \frac{dy}{dx} - 2e^x \\
 -x \sin xy \frac{dy}{dx} - y \sin xy &= 6y \frac{dy}{dx} - 2e^x \\
 (6y + x \sin xy) \frac{dy}{dx} &= 2e^x - y \sin xy \\
 \frac{dy}{dx} &= \frac{2e^x - y \sin xy}{6y + x \sin xy}, (0,1) \\
 &= \frac{2e^0 + \sin(0)}{6 + 0} \\
 &= \frac{1}{3}
 \end{aligned}$$

$$\begin{aligned}
 5) \quad x &= 2t^2 + 1 & y &= 2\sin t^2 \\
 \frac{dx}{dt} &= 4t & \frac{dy}{dt} &= 2\cos t^2 (2t) \\
 & & &= 4t \cos t^2
 \end{aligned}$$

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{dy}{dt} \times \frac{dt}{dx} & \frac{d^2y}{dx^2} &= \frac{d}{dt} \left(\frac{dy}{dx} \right) \left(\frac{dt}{dx} \right) \\
 &= 4t \cos t^2 \left(\frac{1}{4t} \right) & &= -\sin t^2 (2t) \left(\frac{1}{4t} \right) \\
 &= \cos t^2 & &= -\frac{1}{2} \sin t^2
 \end{aligned}$$

$$\begin{aligned}
 6) \quad f(x) &= x^{\frac{1}{4}} + x^{\frac{1}{2}} \\
 f'(x) &= \frac{1}{4x^{\frac{3}{4}}} + \frac{1}{2\sqrt{x}} \\
 x &= 16 \\
 \delta x &= 0.1 \\
 f(x + \delta x) &= f(x) + f'(x) \delta x \\
 &= 6 + \frac{5}{32} \\
 &= 6.1563
 \end{aligned}$$



$$e^{xy} + y^3 = 2x^4 + x^2y$$

$$(x \frac{dy}{dx} + y) e^{xy} + 3y^2 \frac{dy}{dx} = 8x^3 + x^2 \frac{dy}{dx} + 2xy$$

$$u = x \quad u' = 1$$

$$v = x^2 \quad v' = 2x$$

$$v = y \quad v' = \frac{dy}{dx}$$

$$v = y \quad v' = \frac{dy}{dx}$$

$$xe^{xy} \frac{dy}{dx} + 3y^2 \frac{dy}{dx} - x^2 \frac{dy}{dx} = 8x^3 + 2xy - ye^{xy}$$

$$(xe^{xy} + 3y^2 - x^2) \frac{dy}{dx} = 8x^3 + 2xy - ye^{xy}$$

$$\frac{dy}{dx} = \frac{8x^3 + 2xy - ye^{xy}}{xe^{xy} + 3y^2 - x^2}$$

$$ye^{xy} = x^2 + 1$$

$$u = y \quad v' = \frac{dy}{dx}$$

$$v = e^{xy}, \quad v' = x \frac{dy}{dx} + y(e^{xy})$$

$$xye^{xy} \frac{dy}{dx} + y^2 + e^{xy} \frac{dy}{dx} = 2x$$

$$\frac{dy}{dx} (xye^{xy} + e^{xy}) = 2x - y^2 e^{xy}$$

$$\frac{dy}{dx} = \frac{2x - y^2 e^{xy}}{xye^{xy} + e^{xy}}$$

$$\frac{x}{y} - 3x^2 + ye^{x^2} = 1$$

$$u = x \quad u' = 1$$

$$v = y, \quad v' = \frac{dy}{dx}$$

$$v = y \quad v' = \frac{dy}{dx}$$

$$v = e^{x^2}, \quad v' = 2x(e^{x^2})$$

$$\frac{y - x \frac{dy}{dx}}{y^2} - 6x + 2xye^{x^2} + e^{x^2} \frac{dy}{dx} = 0$$

$$y - x \frac{dy}{dx} - 6xy^2 + 2xy^3 e^{x^2} + y^2 e^{x^2} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} (-x + y^2 e^{x^2}) = 6xy^2 - 2xy^3 e^{x^2} - y$$

$$\frac{dy}{dx} = \frac{6xy^2 - y - 2xy^3 e^{x^2}}{y^2 e^{x^2} - x}$$

$$\ln(xy) - 3y^2 = 2x^3$$

$$u = xy, \quad u' = x \frac{dy}{dx} + y$$

$$\frac{x \frac{dy}{dx} + y}{xy} - 6y \frac{dy}{dx} = 6x^2$$

$$x \frac{dy}{dx} + y - 6xy^2 \frac{dy}{dx} = 6x^3 y$$

$$\frac{dy}{dx} (x - 6xy^2) = 6x^3 y - y$$

$$\frac{dy}{dx} = \frac{y(6x^3 - 1)}{x(1 - 6y^2)}$$

$$\ln\left(\frac{x}{y}\right) - 3x^2 + e^{xy} = 1$$

$$v = \ln(u), \quad v' = \frac{1}{u} \times u'$$

$$u = \frac{x}{y}, \quad u' = x \frac{dy}{dx} - y$$

$$b = y \quad b' = \frac{dy}{dx}$$

$$u' = \frac{y - x \frac{dy}{dx}}{y^2}$$

$$\frac{y}{x} \left(\frac{y - x \frac{dy}{dx}}{y^2} \right) - 6x + (x \frac{dy}{dx} + y) e^{xy} = 0$$

$$\frac{y - x \frac{dy}{dx}}{xy} - 6x + xe^{xy} \frac{dy}{dx} + ye^{xy} = 0$$

$$y - x \frac{dy}{dx} - 6x^2 y + x^2 y e^{xy} \frac{dy}{dx} + xy^2 e^{xy} = 0$$

$$\frac{dy}{dx} = \frac{6x^2 y - xy^2 e^{xy} - y}{x^2 y e^{xy} - x}$$

$$\sin(xy) + y \sin x = x$$

$$y = \sin u, u = xy \quad u = y, u' = \frac{dy}{dx}$$

$$\frac{du}{dx} = x \frac{dy}{dx} + y$$

$$v = \sin x, v' = \cos x$$

$$\frac{dy}{du} = \cos u$$

$$\cos xy \left(x \frac{dy}{dx} + y \right) + y \cos x + \frac{dy}{dx} \sin x = 1$$

$$x \cos xy \frac{dy}{dx} + y \cos xy + y \cos x + \sin x \frac{dy}{dx} = 1$$

$$\left(x \cos xy + \sin x \right) \frac{dy}{dx} = 1 - y \cos xy - y \cos x$$

$$\frac{dy}{dx} = \frac{1 - y \cos xy - y \cos x}{x \cos xy + \sin x}$$

$$y = 8^{2x}$$

$$\ln y = \ln 8^{2x}$$

$$\ln y = 2x \ln 8$$

$$\frac{1}{y} \frac{dy}{dx} = 2 \ln 8$$

$$\frac{dy}{dx} = 2(8^{2x}) \ln 8$$

$$y = 2^{\sqrt{x}}$$

$$\ln y = \sqrt{x} \ln 2$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{1}{2\sqrt{x}} \ln 2$$

$$\frac{dy}{dx} = \frac{2^{\sqrt{x}}}{2\sqrt{x}} \ln 2$$

$$y = 3^{x^2 \ln x}$$

$$\ln y = \ln 3^{x^2 \ln x}$$

$$\ln y = x^2 \ln x \ln 3$$

$$u = x^2 \quad u' = 2x$$

$$v = \ln x \quad v' = \frac{1}{x}$$

$$\frac{1}{y} \frac{dy}{dx} = x + 2x \ln x \ln 3$$

$$\frac{dy}{dx} = xy \ln 3 + 2xy \ln x \ln 3$$

$$= x 3^{x^2 \ln x} + 2x (3^{x^2 \ln x}) (\ln x) (\ln 3)$$

$$y = 2^{5x} \times 3^{4x^2}$$

$$\ln y = \ln (2^{5x} \times 3^{4x^2})$$

$$\ln y = \ln 2^{5x} + \ln 3^{4x^2}$$

$$\frac{1}{y} \frac{dy}{dx} = 5 \ln 2 + 8x \ln 3$$

$$\frac{dy}{dx} = 5y \ln 2 + 8xy \ln 3$$

$$= 5(2^{5x} \cdot 3^{4x^2}) (\ln 2)$$

$$+ 8x (2^{5x} \cdot 3^{4x^2}) (\ln 3)$$

LIST OF FORMULA

Trigonometric
$\cos^2 x + \sin^2 x = 1$ $1 + \tan^2 x = \sec^2 x$ $\cot^2 x + 1 = \operatorname{cosec}^2 x$ $\sin 2x = 2 \sin x \cos x$ $\cos 2x = \cos^2 x - \sin^2 x$ $= 2 \cos^2 x - 1$ $= 1 - 2 \sin^2 x$ $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$ $\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$ $\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$ $\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$ $2 \sin x \cos y = \sin(x + y) + \sin(x - y)$ $2 \sin x \sin y = -\cos(x + y) + \cos(x - y)$ $2 \cos x \cos y = \cos(x + y) + \cos(x - y)$
Logarithm
$a^x = e^{x \ln a}$ $\log_a x = \frac{\log_b x}{\log_b a}$

Differentiations
$\frac{d}{dx}[k] = 0, \quad k \text{ constant}$
$\frac{d}{dx}[x^n] = nx^{n-1}$
$\frac{d}{dx}[e^x] = e^x$
$\frac{d}{dx}[\ln x] = \frac{1}{x}$
$\frac{d}{dx}[\cos x] = -\sin x$
$\frac{d}{dx}[\sin x] = \cos x$
$\frac{d}{dx}[\tan x] = \sec^2 x$
$\frac{d}{dx}[\cot x] = -\operatorname{cosec}^2 x$
$\frac{d}{dx}[\sec x] = \sec x \tan x$
$\frac{d}{dx}[\operatorname{cosec} x] = -\operatorname{cosec} x \cot x$